

ATHARVA R ANKAD

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EDUCATION

VIT Bhopal University <i>Integrated M.Tech in Computer Science and Engineering (Computational and Data Science)</i>	Bhopal, India 2023-2028
Sri Chaitanya Techno School <i>CBSE Class XII Science</i>	Bangalore, India 2021-2023
HAL Gnanajyoti School <i>ICSE Class X</i>	Bangalore, India 2009-2021

TECHNICAL SKILLS

Languages: Python, Java, SQL
ML / AI: Machine Learning, NLP, Retrieval-Augmented Generation (RAG), Transformers, scikit-learn, SentenceTransformers, LangChain
Backend & APIs: FastAPI, REST API Design
Data & Libraries: Pandas, NumPy, MongoDB, Hadoop (Fundamentals)
Tools & Platforms: Git, Linux
Concepts: Data Engineering, Data Integration, Vector Search, Pipeline Design

PROJECTS

Fasal Sahayak AI <i>Python, FastAPI, scikit-learn, RandomForest, Pandas, NASA POWER API</i>	GitHub
- Built an end-to-end ML pipeline and REST API for data-driven crop yield prediction and agronomic advisory, integrating 13 heterogeneous data sources including 16K+ historical crop records, live meteorological data (NASA POWER API), and soil research from ICAR and government publications. - Designed a two-stage RandomForestRegressor pipeline separating outlier crops (e.g., sugarcane) from annual crops, applying log1p target normalization to handle skewed yield distributions. - Developed a dual-mode advisory engine providing generalized agronomic guidance for standard users and threshold-based warnings for farmers entering Soil Health Card (SHC) metrics. - Reduced query latency by implementing lightweight JSON lookup stores for pre-computed geographic data, eliminating runtime database calls.	
RAG Semantic Search System <i>Python, LangChain, SentenceTransformers, MongoDB, Groq API</i>	GitHub
- Built a high-performance Retrieval-Augmented Generation (RAG) pipeline that ingests PDF documents, generates local semantic embeddings using SentenceTransformers (all-MiniLM-L6-v2), and performs vector similarity search via MongoDB Atlas Vector Search. - Implemented context-aware smart chunking with LangChain text splitters to preserve paragraph structure and semantic meaning across document chunks. - Integrated Groq API with a strict contextual system prompt to ensure LLM responses are grounded solely in retrieved document context, with citation support including page numbers and relevance scores.	

ACHIEVEMENTS

Smart India Hackathon (SIH) 2025 — Internal Nominee <i>Selected among the top 45 out of 949 idea submissions in the university's internal hackathon</i>	VIT Bhopal University 2024
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CERTIFICATIONS

Google IT Support Certificate <i>Foundations of IT support, networking, and systems administration</i>	Google 2026
Building RAG Apps using MongoDB <i>Credential in Retrieval-Augmented Generation application development</i>	MongoDB 2026